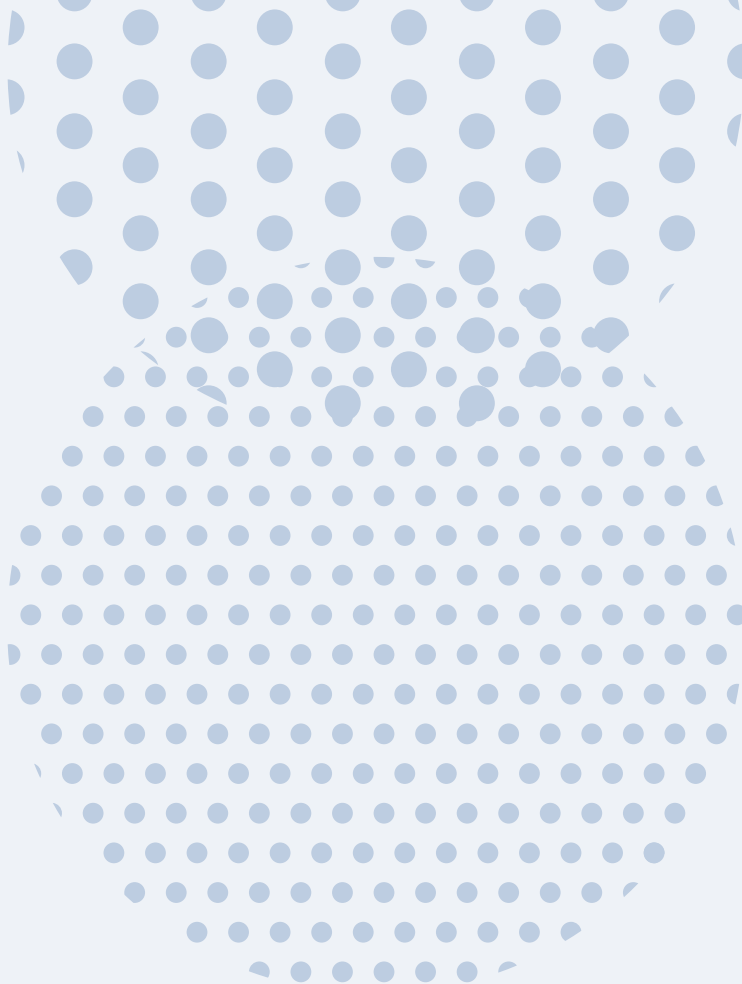


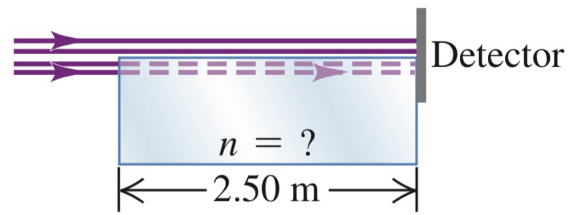


**33.6** ●● Light of a certain frequency has a wavelength of 526 nm in water. What is the wavelength of this light in benzene?

**33.8 ••** A laser beam shines along the surface of a block of transparent material (see **Fig. E33.8**). Half of the beam goes straight to a detector, while the other half travels through the block and then hits the detector.

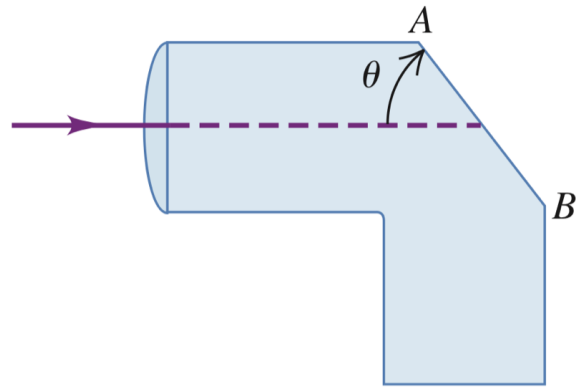
The time delay between the arrival of the two light beams at the detector is 6.15 ns. What is the index of refraction of this material?

Figure **E33.8**



**33.15 • Light Pipe.** Light enters a solid pipe made of plastic having an index of refraction of 1.60. The light travels parallel to the upper part of the pipe (**Fig. E33.15**). You want to cut the face  $AB$  so that all the light will reflect back into the pipe after it first strikes that face. (a) What is the largest that  $\theta$  can be if the pipe is in air? (b) If the pipe is immersed in water of refractive index 1.33, what is the largest that  $\theta$  can be?

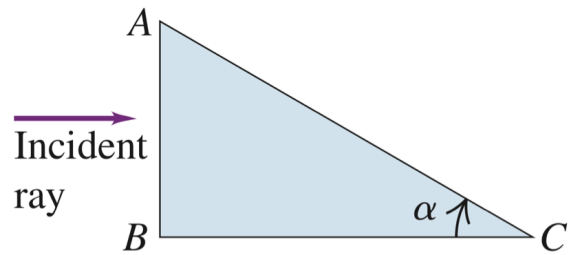
Figure **E33.15**





**33.21 •** Light is incident along the normal on face  $AB$  of a glass prism of refractive index 1.52, as shown in **Fig. E33.21**. Find the largest value the angle  $\alpha$  can have without any light refracted out of the prism at face  $AC$  if (a) the prism is immersed in air and (b) the prism is immersed in water.

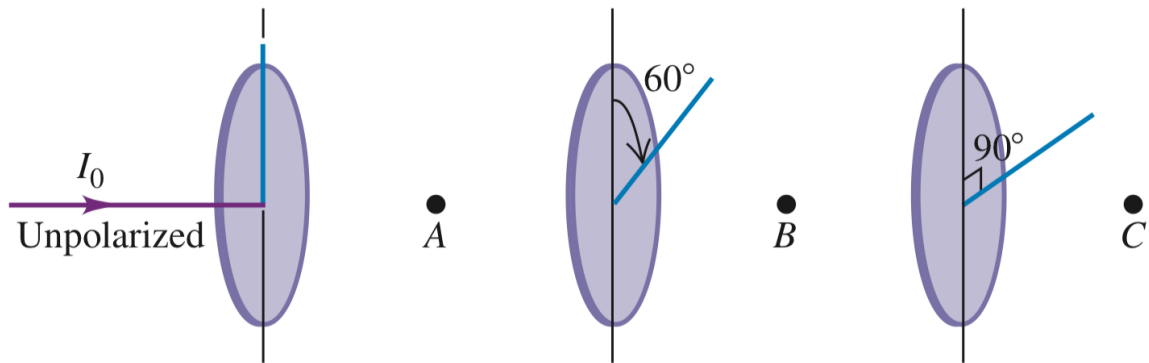
Figure **E33.21**



**33.28 •** A layer of liquid sits on top of the horizontal surface of a transparent solid. For a ray traveling in the solid and incident on the interface of the two materials, the critical angle is  $38.7^\circ$ . (a) For a ray traveling in the solid and reflecting at the interface with the liquid, for what incident angle with respect to the normal is the reflected ray 100% polarized? (b) What is the polarizing angle if the ray is traveling in the liquid?

**33.29 ••** A beam of unpolarized light of intensity  $I_0$  passes through a series of ideal polarizing filters with their polarizing axes turned to various angles as shown in **Fig. E33.29**. (a) What is the light intensity (in terms of  $I_0$ ) at points  $A$ ,  $B$ , and  $C$ ? (b) If we remove the middle filter, what will be the light intensity at point  $C$ ?

Figure **E33.29**

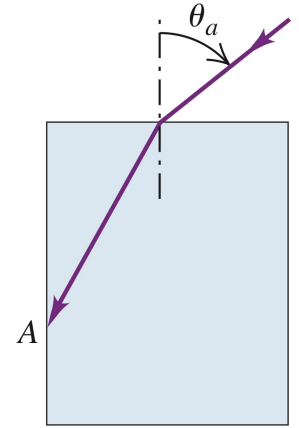


**33.34** ●●● Three polarizing filters are stacked, with the polarizing axis of the second and third filters at  $23.0^\circ$  and  $62.0^\circ$ , respectively, clockwise to that of the first. If unpolarized light is incident on the stack, the light has intensity  $55.0 \text{ W/cm}^2$  after it passes through the stack. If the incident intensity is kept constant but the second polarizer is removed, what is the intensity of the light after it has passed through the stack?

**33.38** ●●● In a physics lab, light with wavelength 530 nm travels in air from a laser to a photocell in 16.7 ns. When a slab of glass 0.850 m thick is placed in the light beam, with the beam incident along the normal to the parallel faces of the slab, it takes the light 20.8 ns to travel from the laser to the photocell. What is the wavelength of the light in the glass?

Figure P33.39

**33.39** •• A ray of light is incident in air on a block of a transparent solid whose index of refraction is  $n$ . If  $n = 1.37$ , what is the *largest* angle of incidence  $\theta_a$  for which total internal reflection will occur at the vertical face (point  $A$  shown in **Fig. P33.39**)?



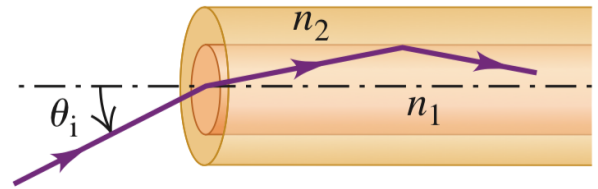
**33.41** ●● A ray of light traveling *in* a block of glass ( $n = 1.52$ ) is incident on the top surface at an angle of  $57.2^\circ$  with respect to the normal in the glass. If a layer of oil is placed on the top surface of the glass, the ray is totally reflected. What is the maximum possible index of refraction of the oil?

**33.43** ●●● A glass plate 2.65 mm thick, with an index of refraction of 1.80, is placed between a point source of light with wavelength 600 nm (in vacuum) and a screen. The distance from source to screen is 1.30 cm. How many wavelengths are there between the source and the screen?



**33.46 ••** Optical fibers are constructed with a cylindrical core surrounded by a sheath of cladding material. Common materials used are pure silica ( $n_2 = 1.450$ ) for the cladding and silica doped with germanium ( $n_1 = 1.465$ ) for the core. (a) What is the critical angle  $\theta_{\text{crit}}$  for light traveling in the core and reflecting at the interface with the cladding material? (b) The numerical aperture (NA) is defined as the angle of incidence  $\theta_i$  at the flat end of the cable for which light is incident on the core–cladding interface at angle  $\theta_{\text{crit}}$  (**Fig. P33.46**). Show that  $\sin \theta_i = \sqrt{n_1^2 - n_2^2}$ . (c) What is the value of  $\theta_i$  for  $n_1 = 1.465$  and  $n_2 = 1.450$ ?

Figure **P33.46**



**33.48 ••** When linearly polarized light passes through a polarizer, its polarizing axis may be rotated by any angle  $\phi < 90^\circ$  at the expense of a loss of intensity, as determined by Malus's law. By using sequential polarizers, you can achieve a similar axis rotation but retain greater intensity. In fact, if you use many intermediate polarizers, the polarization axis can be rotated by  $90^\circ$  with virtually undiminished intensity.

(a) Derive an equation for the resulting intensity if linearly polarized light passes through successive  $N$  polarizers, each with the polarizing axis rotated by an angle  $90^\circ/2N$  larger than the preceding polarizer.

(b) By making a table of the resulting intensity for various values of  $N$ , estimate the minimum number  $N$  of polarizers needed so that the light will have its polarization axis rotated by  $90^\circ$  while maintaining more than 90% of its intensity.

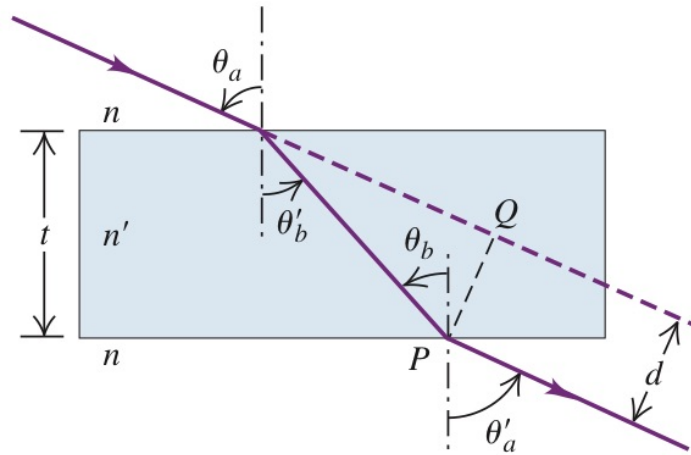
(c) Estimate the minimum number of polarizers needed to maintain more than 95% and 99% intensity.

**33.52 ••** Light is incident in air at an angle  $\theta_a$  (**Fig. P33.52**) on the upper surface of a transparent plate, the surfaces of the plate being plane and parallel to each other. (a) Prove that  $\theta_a = \theta'_a$ . (b) Show that this is true for any number of different parallel plates. (c) Prove that the lateral displacement  $d$  of the emergent beam is given by the relationship

$$d = t \frac{\sin(\theta_a - \theta'_b)}{\cos \theta'}$$

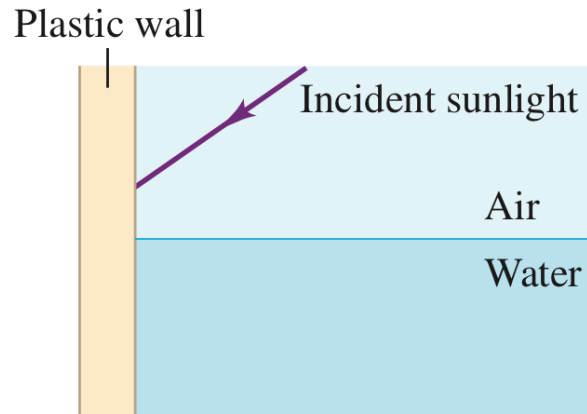
where  $t$  is the thickness of the plate. (d) A ray of light is incident at an angle of  $66.0^\circ$  on one surface of a glass plate 2.40 cm thick with an index of refraction of 1.80. The medium on either side of the plate is air. Find the lateral displacement between the incident and emergent rays.

Figure **P33.52**



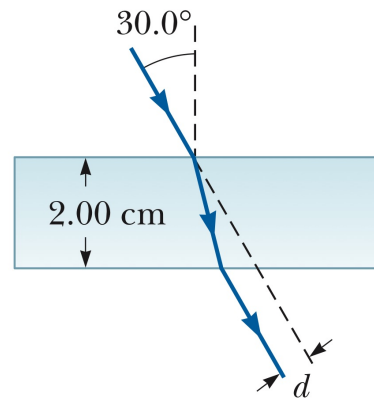
**33.53** ●● A beam of unpolarized sunlight strikes the vertical plastic wall of a water tank at an unknown angle. Some of the light reflects from the wall and enters the water (**Fig. P33.53**). The refractive index of the plastic wall is 1.67. If the light that has been reflected from the wall into the water is observed to be completely polarized, what angle does this beam make with the normal inside the water?

Figure **P33.53**



- 19.** When you look through a window, by what time interval is the light you see delayed by having to go through glass instead of air? Make an order-of-magnitude estimate on the basis of data you specify. By how many wavelengths is it delayed?

- 22.** When the light ray illustrated in Figure P35.22 passes through the glass block of index of refraction  $n = 1.50$ , it is shifted laterally by the distance  $d$ . (a) Find the value of  $d$ . (b) Find the time interval required for the light to pass through the glass block.



**Figure P35.22**

Figure P35.26

- 27.** An opaque cylindrical tank with an open top has a diameter of 3.00 m and is completely filled with water. When the afternoon Sun reaches an angle of  $28.0^\circ$  above the horizon, sunlight ceases to illuminate any part of the bottom of the tank. How deep is the tank?

32. A person looking into an empty container is able to see the far edge of the container's bottom as shown in Figure P35.32a. The height of the container is  $h$ , and its width is  $d$ . When the container is completely filled with a fluid of index of refraction  $n$  and viewed from the same angle, the person can see the center of a coin at

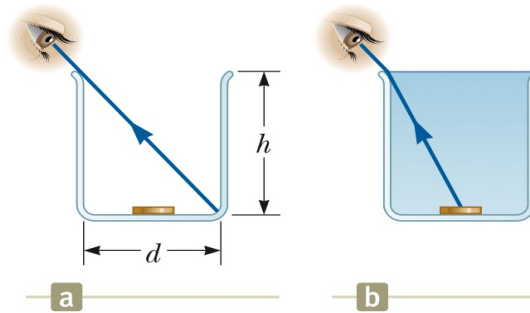


Figure P35.32

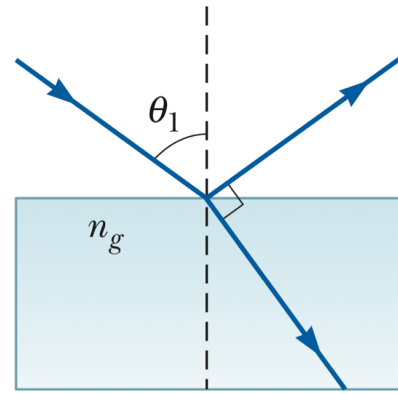
the middle of the container's bottom as shown in Figure P35.32b. (a) Show that the ratio  $h/d$  is given by

$$\frac{h}{d} = \sqrt{\frac{n^2 - 1}{4 - n^2}}$$

(b) Assuming the container has a width of 8.00 cm and is filled with water, use the expression above to find the height of the container. (c) For what range of values of  $n$  will the center of the coin not be visible for any values of  $h$  and  $d$ ?

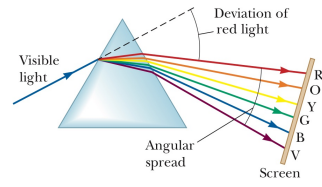


- 35.** A beam of light both reflects and refracts at the surface between air and glass as shown in Figure P35.35. If the refractive index of the glass is  $n_g$ , find the angle of incidence  $\theta_1$  in the air that would result in the reflected ray and the refracted ray being perpendicular to each other.



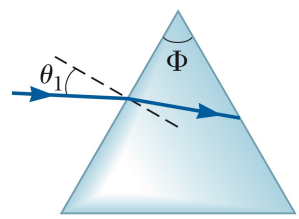
**Figure P35.35**

See Figure P35.39.



**Figure P35.39** Problems 39 and 40.

40. The index of refraction for violet light in silica flint glass is  $n_v$ , and that for red light is  $n_r$ . What is the angular spread of visible light passing through a prism of apex angle  $\Phi$  if the angle of incidence is  $\theta$ ? See Figure P35.39.



**Figure P35.43**  
Problems 43 and 44.

- 44.** A triangular glass prism with apex angle  $\Phi$  has an index of refraction  $n$  (Fig. P35.43). What is the smallest angle of incidence  $\theta_1$  for which a light ray can emerge from the other side?

Problems 43 and 44.

45. Assume a transparent rod of diameter  $d = 2.00 \mu\text{m}$  has an index of refraction of 1.36. Determine the maximum angle  $\theta$  for which the light rays incident on the end of the rod in Figure P35.45 are subject to total internal reflection along the walls of the rod. Your answer defines the size of the *cone of acceptance* for the rod.

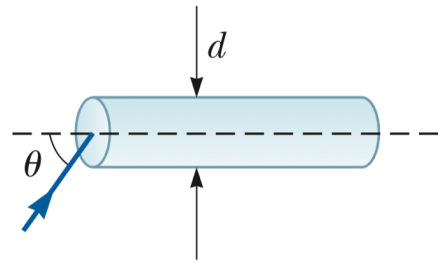


Figure P35.45

49. An optical fiber has an index of refraction  $n$  and diameter  $d$ . It is surrounded by vacuum. Light is sent into the fiber along its axis as shown in Figure P35.49. (a) Find the smallest outside radius  $R_{\min}$  permitted for a bend in the fiber if no light is to escape. (b) **What If?** What result does part (a) predict as  $d$  approaches zero? Is this behavior reasonable? Explain. (c) As  $n$  increases? (d) As  $n$  approaches 1? (e) Evaluate  $R_{\min}$  assuming the fiber diameter is  $100\ \mu\text{m}$  and its index of refraction is 1.40.

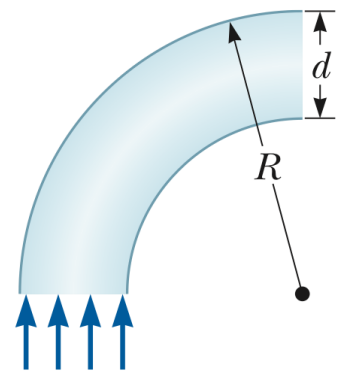


Figure P35.49

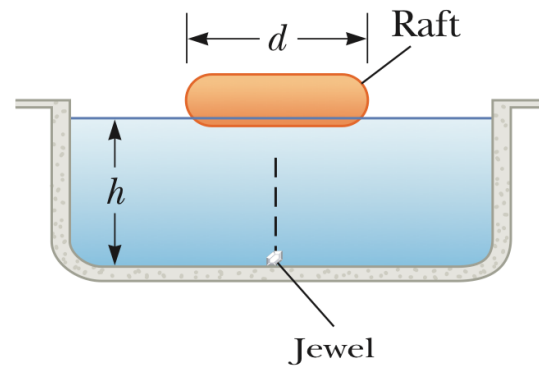
**57.** When light is incident normally on the interface between two transparent optical media, the intensity of the reflected light is given by the expression

$$S_1' = \left( \frac{n_2 - n_1}{n_2 + n_1} \right)^2 S_1$$

In this equation,  $S_1$  represents the average magnitude of the Poynting vector in the incident light (the incident intensity),  $S_1'$  is the reflected intensity, and  $n_1$  and  $n_2$  are the refractive indices of the two media. (a) What fraction of the incident intensity is reflected for 589-nm light normally incident on an interface between air and crown glass? (b) Does it matter in part (a) whether the light is in the air or in the glass as it strikes the interface?

**58.** Refer to Problem 57 for its description of the reflected intensity of light normally incident on an interface between two transparent media. (a) For light normally incident on an interface between vacuum and a transparent medium of index  $n$ , show that the intensity  $S_2$  of the transmitted light is given by  $S_2/S_1 = 4n/(n + 1)^2$ . (b) Light travels perpendicularly through a diamond slab, surrounded by air, with parallel surfaces of entry and exit. Apply the transmission fraction in part (a) to find the approximate overall transmission through the slab of diamond, as a percentage. Ignore light reflected back and forth within the slab.

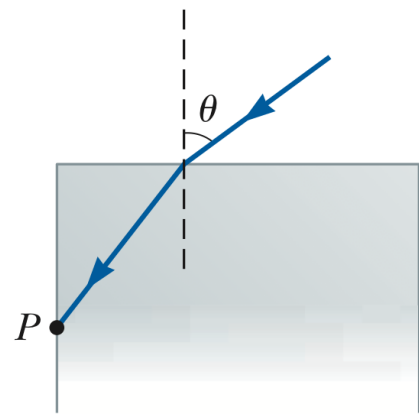
- 63.** A thief hides a precious jewel by placing it on the bottom of a public swimming pool. He places a circular raft on the surface of the water directly above and centered over the jewel as shown in Figure P35.63. The surface of the water is calm. The raft, of diameter  $d = 4.54$  m, prevents the jewel from being seen by any observer above the water, either on the raft or on the side of the pool. What is the maximum depth  $h$  of the pool for the jewel to remain unseen?



**Figure P35.63**



- 68.** A light ray of wavelength 589 nm is incident at an angle  $\theta$  on the top surface of a block of polystyrene as shown in Figure P35.68. (a) Find the maximum value of  $\theta$  for which the refracted ray undergoes total internal reflection at the point  $P$  located at the left vertical face of the block. **What If?** Repeat the calculation for the case in which the polystyrene block is immersed in (b) water and (c) carbon disulfide. Explain your answers.



**Figure P35.68**

76. A. H. Pfund's method for measuring the index of refraction of glass is illustrated in Figure P35.76. One face of a slab of thickness  $t$  is painted white, and a small hole scraped clear at point  $P$  serves as a source of diverging rays when the slab is illuminated from below. Ray  $PBB'$  strikes the clear surface at the critical angle and is totally reflected, as are rays such as  $PCC'$ . Rays such as  $PAA'$  emerge from the clear surface. On the painted surface, there appears a dark circle of diameter  $d$  surrounded by an illuminated region, or halo. (a) Derive an equation for  $n$  in terms of the measured quantities  $d$  and  $t$ . (b) What is the diameter of the dark circle if  $n = 1.52$  for a slab 0.600 cm thick? (c) If white light is used, dispersion causes the critical angle to depend on color. Is the inner edge of the white halo tinged with red light or with violet light? Explain.

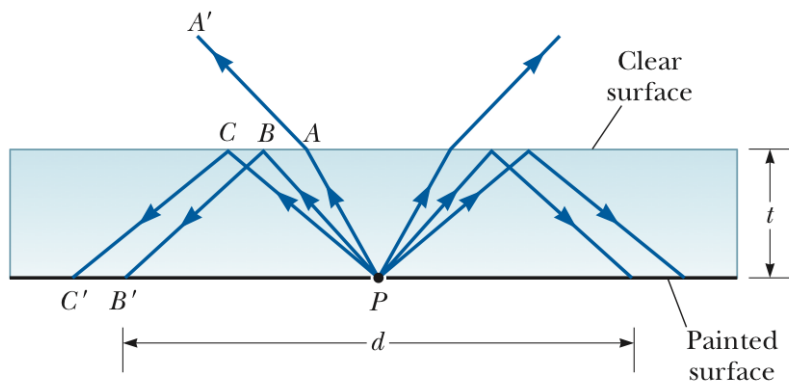
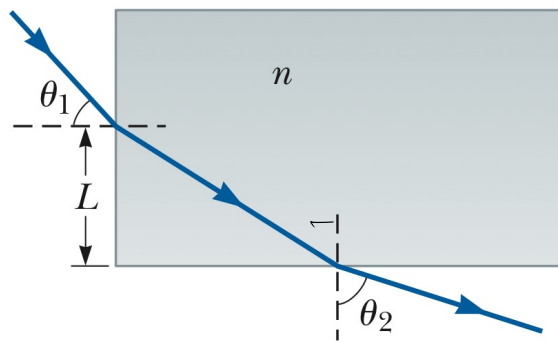


Figure P35.76

- 77.** A light ray enters a rectangular block of plastic at an angle  $\theta_1 = 45.0^\circ$  and emerges at an angle  $\theta_2 = 76.0^\circ$  as shown in Figure P35.77. (a) Determine the index of refraction of the plastic. (b) If the light ray enters the plastic at a point  $L = 50.0$  cm from the bottom edge, what time interval is required for the light ray to travel through the plastic?



**Figure P35.77**

84. Pierre de Fermat (1601–1665) showed that whenever light travels from one point to another, its actual path is the path that requires the smallest time interval. This statement is known as *Fermat's principle*. The simplest example is for light propagating in a homogeneous medium. It moves in a straight line because a straight line is the shortest distance between two points. Derive Snell's law of refraction from Fermat's principle. Proceed as follows. In Figure P35.84, a light ray travels from point  $P$  in medium 1 to point  $Q$  in medium 2. The two points are, respectively, at perpendicular distances  $a$  and  $b$  from the interface. The displacement from  $P$  to  $Q$  has the component  $d$  parallel to the interface, and we let  $x$  represent the coordinate of the point where the ray enters the second medium. Let  $t = 0$  be the instant the light starts from  $P$ . (a) Show that the time at which the light arrives at  $Q$  is

$$t = \frac{r_1}{v_1} + \frac{r_2}{v_2} = \frac{n_1\sqrt{a^2 + x^2}}{c} + \frac{n_2\sqrt{b^2 + (d - x)^2}}{c}$$

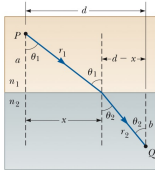


Figure P35.84 Problems 84 and 85.

85. Refer to Problem 84 for the statement of Fermat's principle of least time. Derive the law of reflection (Eq. 35.2) from Fermat's principle.

- 86.** Suppose a luminous sphere of radius  $R_1$  (such as the Sun) is surrounded by a uniform atmosphere of radius  $R_2 > R_1$  and index of refraction  $n$ . When the sphere is viewed from a location far away in vacuum, what is its apparent radius (a) when  $R_2 > nR_1$  and (b) when  $R_2 < nR_1$ ?